From acceleration to velocity

Source slide - academic English

MOTION / ORIGINAL PUBLIC LESSON

From acceleration to velocity

Constant acceleration produces equal velocity changes over equal time intervals.

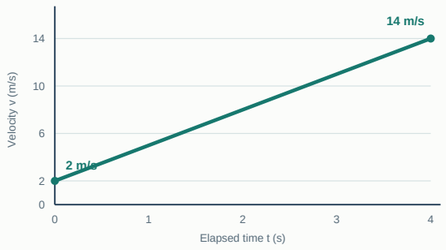
Initial velocity: 2 m/s

Constant acceleration: 3 m/s per second

Observation interval: 0 to 4 s

$$v(t) = v_0 + a\,t$$

v0 = initial velocity a = acceleration t = time



The line's slope is acceleration; its starting height is initial velocity.

Constant acceleration / v0 = 2 m/s, a = 3 m/s2, t = 4 s 01 / 03

Mirror slide - plain English

MOTION / ORIGINAL PUBLIC LESSON

From acceleration to velocity

When acceleration stays the same, velocity changes by the same amount each second.

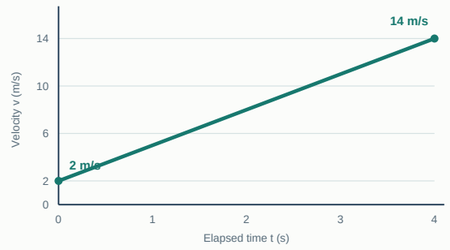
Start at a velocity of 2 m/s

Add 3 m/s of velocity every second

Watch the first 4 seconds of motion

$$v(t) = v_0 + a\,t$$

v0 = initial velocity a = acceleration t = time



The slope tells us how velocity changes; the line starts at the initial velocity.

Constant acceleration / v0 = 2 m/s, a = 3 m/s2, t = 4 s 01 / 03

Vocabulary

Initial	At the start; v0 is velocity at time zero.
Rate	Change per unit of time.
Slope	Change in graph height divided by change in time.

Prerequisite bridge: A subscript identifies a particular value. The 0 in v0 labels the start; it does not multiply v. Acceleration of 3 m/s2 adds 3 m/s of velocity each second.

Homework connection: For part (a), first calculate the gain a t = 12 m/s, then add the starting 2 m/s. The endpoint 14 m/s also feeds the graph check in part (c). (Full authored question and guided solution: study-guide pages 6-7.)

Original authored demo / slide 1

Study-guide pages 1-2, 6

Displacement is the area under velocity

Source slide - academic English

MOTION / ORIGINAL PUBLIC LESSON

Displacement is the area under velocity

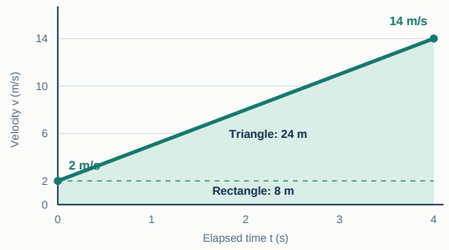
For this motion, the velocity-time region is a trapezoid.
Its area equals the displacement over the interval.

Average velocity is the mean of the endpoints.
Multiply average velocity by elapsed time.

$$\bar{v} = \frac{v_0 + v(t)}{2}$$

$$\Delta x = \bar{v}\, t$$

v-bar = average velocity delta x = displacement



The graph converts a velocity calculation into a position change.

Constant acceleration / v0 = 2 m/s, a = 3 m/s2, t = 4 s 02 / 03

Mirror slide - plain English

MOTION / ORIGINAL PUBLIC LESSON

Displacement is the area under velocity

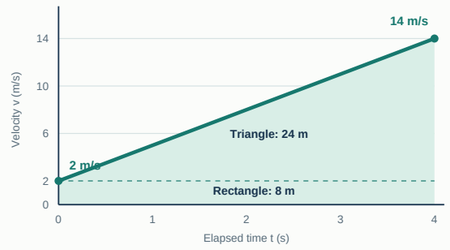
The space below this velocity line is a trapezoid.
Its area tells us how far the cart moves forward.

Average velocity is halfway between start and finish.
Multiply that average by the number of seconds.

$$\bar{v} = \frac{v_0 + v(t)}{2}$$

$$\Delta x = \bar{v}\, t$$

v-bar = average velocity delta x = displacement



The graph links how fast the cart moves to how its position changes.

Constant acceleration / v0 = 2 m/s, a = 3 m/s2, t = 4 s 02 / 03

Vocabulary

Displacement	Final position minus starting position; it is signed.
Average velocity	The constant velocity producing the same displacement.
Trapezoid	A four-sided region with one pair of parallel sides.

Why this relation follows: Velocity times a short time interval gives a small displacement. Add these strips to get the area. Since the velocity line is straight, its average height is halfway between the endpoint heights.

Homework connection: Part (c) splits the shaded region into an 8 m rectangle and a 24 m triangle. Their sum checks part (b). Distance equals forward displacement here because the cart does not reverse. (Original authored graph; study-guide pages 3 and 7.)

Original authored demo / slide 2

Study-guide pages 3, 7

Build the constant-acceleration equation

Source slide - academic English

MOTION / ORIGINAL PUBLIC LESSON

Build the constant-acceleration equation

Substitute the velocity expression into the area relation.
Distribute the elapsed-time factor over both terms.

$$\Delta x = \frac{v_0 + (v_0 + a\,t)}{2}\,t$$

$$\Delta x = v_0\,t + \tfrac{1}{2}a\,t^2$$

Existing motion contributes the rectangular area.

Acceleration contributes the additional triangular area.

delta x = position change v0 = initial velocity a = acceleration t = elapsed time

Every term has units of length; this model requires constant acceleration.

Constant acceleration / v0 = 2 m/s, a = 3 m/s2, t = 4 s 03 / 03

Mirror slide - plain English

MOTION / ORIGINAL PUBLIC LESSON

Build the constant-acceleration equation

Put the velocity formula into the formula for the area.
Then multiply time into each part of the expression.

$$\Delta x = \frac{v_0 + (v_0 + a\,t)}{2}\,t$$

$$\Delta x = v_0\,t + \tfrac{1}{2}a\,t^2$$

The starting velocity gives the rectangle's area.

The extra velocity gives the triangle's area.

delta x = position change v0 = initial velocity a = acceleration t = elapsed time

Both parts give metres; use this formula only while acceleration stays constant.

Constant acceleration / v0 = 2 m/s, a = 3 m/s2, t = 4 s 03 / 03

Vocabulary

Substitution	Replace a quantity with an expression equal to it.
Distribute	Multiply every term inside a sum by the outside factor.
Dimension	The physical kind of a quantity, such as length or time.

The new notation and the one-half: Delta x is change in position; t squared is time multiplied by itself. The one-half comes from the triangular extra area, not from an arbitrary instruction to divide the final answer.

Homework connection: Part (b) separates starting-motion displacement, 8 m, from acceleration's extra 24 m. Both terms must have units of metres. The new practice on page 8 checks whether the reasoning transfers. (All lesson text and questions are independently authored for this repository.)